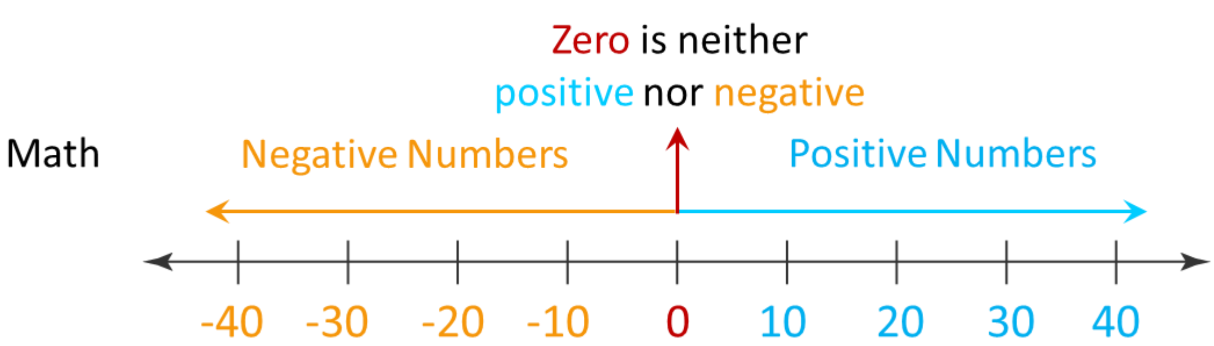
How to Converse Accurately About Matric Potential

Matric potential is a special case of pressure potential in which water molecules in a porous medium, such as soil, develop a negative pressure below atmospheric pressure due to adhesive intermolecular and/or capillary forces. Matric potential can be thought of as a resistive or pulling force, conventionally reported as unsigned magnitudes such as "tension", "suction", or "vacuum". This pulling force is measured using negative numbers, with more negative values indicating stronger tension. As such, we must think differently, "in reverse," as it were, to converse accurately about matric potential. But first, let's go back to school for a refresher about numbers.

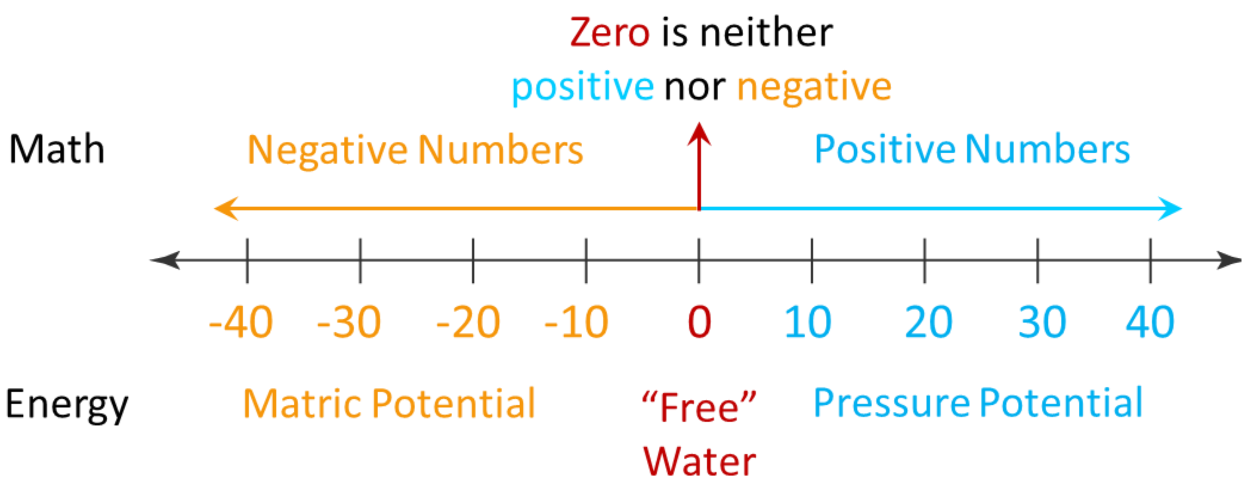
In grade-school mathematics, "real" numbers were represented as positive or negative values. A graphic was devised to help visualize the representation of numbers on a straight line:



We can observe that the numbers increase or decrease as we change direction along the line. By definition, any number to the **right** is always **more than** the number to its **left**. For example, 40 is **greater than** 10; -10 is **greater than** -40.

Similarly, any number to the **left** is always **less than** the number to its **right**. For example, 10 is **less than** 40; -40 is **less than** -10. The same rule applies to numbers represented as integers, fractions, or radicals (√) and exponents (a^b) that can be expressed as rational numbers.

In physics, energy is a quantitative property of matter that can be stored, transferred, or used to perform work. Water, as a substance, also has energy that depends on its state. The potential energy of water can be represented visually by the same number line:



Where positive numbers represent water with potential free energy (pressure potential) available to move (i.e., do work) and negative numbers represent water that is under tension with less free energy to move (matric potential). Water at zero potential is pure, at standard temperature, and not in a place where it is pushed or pulled by anything. This resting place is conventionally called "free" water.

We observe the same rule applied to numbers that quantify energy, using slightly different jargon: any number to the **right** always has **more** potential free energy than the number to its **left**. For example, 40 psi is greater than 10 psi; thus, water at 40 psi has more potential free energy than at 10 psi. As we move to the left, crossing the zero point puts us in negative territory, but the rule still applies: the free energy of water at -10 matric potential (-1.45 psi) is greater than at -40 matric potential (-5.8 psi), i.e., it has **more** freedom of movement. Conversely, a matric potential of -40 kPa is less than -10 kPa, indicating **less** freedom of movement.

In writing about matric potential, the following relations are observed:

Higher matric potential = **Less** negative value, i.e., closer to zero potential
Lower matric potential = **More** negative value, i.e., further away (left) from zero potential

Matric potential **decreases** as it becomes **more** negative.
Matric potential **increases** as it becomes **less** negative.
If matric potential changes from -10 to -40, then it's **decreasing,** i.e., becoming more negative.
If matric potential changes from -40 to -10, then it's **increasing,** i.e., becoming less negative.

Water always moves from an area of high potential to an area of low potential, regardless of the sign of the potential difference. Possible units of matric potential are the Pascal (Pa, SI), bar (metric), energy per unit mass (thermodynamic), or pounds per square inch (psi, Imperial). In writing, matric potential is always preceded by a negative (-) sign. Possible units of pressure potential are the Pascal (Pa, SI); centimeters of water (cm H2O, manometric); millimeters of mercury (mm Hg, manometric), and pounds per square inch (psi, Imperial). In writing, units of positive pressure are unsigned.

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Dirt Hog's Companion